Speech Science - Lecture 01

Lecture overview

Lecture 01 introduces Speech Science 1, focusing on the distinction between the physical continuity of speech signals and the psychological perception of discrete words, a duality that challenges computer processing systems. The course covers phonetics as the language-independent study of speech production, acoustics, and perception, contrasting it with phonology, which analyzes language-specific sound patterns. Key physiological mechanisms include the pulmonic egressive airstream, where air from the lungs is modulated by vocal fold vibration (phonation) and vocal tract constrictions to generate voiced, voiceless, fricative, and plosive sounds. The lecture details how fundamental frequency (F0) determines pitch through vocal fold tension and mass, while intensity relates to airflow resistance and amplitude.

The session further explores the classification of speech sounds using parameters such as manner, place, and voicing, alongside the International Phonetic Alphabet (IPA) for unambiguous transcription. A critical distinction is made between phonemes, the smallest meaningful contrastive units in a language, and phones, their physical realizations or allophones. The lecture emphasizes that many phonetic features, such as aspiration in English or nasalization before nasal consonants, are redundant and predictable by rule rather than distinctive, unlike in languages like Thai where these features alter meaning. Understanding these systematic variations and the rules governing permissible sound sequences is essential for distinguishing natural speech from synthetic output and for accurate linguistic analysis.

Slide 1: Speech Science 1 2026-2027

This slide introduces the "Speech Science 1" course for the 2026-2027 academic year. The lecture is presented by Professor Astrid van Wieringen from the Department of Neurosciences at KU Leuven, specifically within the Experimental ORL division. No additional content or explanation accompanies this introductory title slide in the provided sources.

Slide 2: • Classes: see Toledo (and watch out for changes)

Students must consult Toledo for class schedules while monitoring for updates. Course content includes five to six PowerPoint presentations, Praat exercises, and two papers. The closed-book exam emphasizes conceptual understanding over rote memorization; SLT students submit blok exercises three days prior, while DH students additionally complete a two-point assignment.

Slide 3: • What do you hear when you listen to your own language?

When listening to one's own language, listeners perceive discrete words with distinct meanings, whereas a foreign language often sounds like a continuous stream where word boundaries are unclear. This creates a dual nature of speech: physically, the signal is continuous, but psychologically, it is processed as discrete groups of sounds. This discrepancy between the physical continuity and psychological discreteness presents a significant challenge for computer processing systems attempting to replicate human speech understanding.

Slide 4: Phonetics: the science of speech sounds (language independent.)

Phonetics is the language-independent science of speech sounds, encompassing articulatory (production), acoustic (physical properties like voicing and frication), and auditory (perception) subfields. Unlike phonology, which analyzes language-dependent sound patterns, phonetics treats speech as a continuous physical waveform generated by the lungs, vocal folds, and vocal tract. Understanding these fundamental physical mechanisms is essential for programming computers to process speech, as machines lack the innate human capacity to acquire language.

Slide 5: • Difficult not to think of language: From our linguistic knowledge

Our linguistic knowledge allows us to distinguish meaningful sounds from unimportant ones, such as clicks or coughs, and to segment continuous speech into individual words. This ability explains why we perceive foreign languages as having "run together" words due to a lack of audible breaks, even though native speakers also produce speech without distinct pauses between words.

Slide 6: Text to Speech and Speech to Text

Slide 6 introduces the concepts of Text to Speech and Speech to Text. The provided source material consists solely of this title, with no accompanying definitions, technical details, examples, or professor explanations available in the aligned transcript or retrieval context. Consequently, no further elaboration on these topics can be derived from the supplied information.

Slide 7: Speech production

Speaking is a motor skill where air moves from the lungs through the trachea and pharynx to oral and nasal cavities. Speech sounds arise when vocal tract constrictions impede airflow, causing pressure variations in the trachea. This process involves continuous alternation between constricted and open articulations without gaps, occurring at an average rate of 2-5 syllables per second depending on speaking style.

Slide 8: From English Phonetics and phonology, P. Roach

The larynx houses the vocal folds. Their vibration produces voicing; when the folds do not vibrate in the same way, the sound can be voiceless. This connects anatomy to the voiced-versus-voiceless distinction heard in speech.

Slide 9: The larynx sits on top of trachea, a series of horseshoe-shaped cartilages, open at the back (cartilage = kraakbeen). It controls the flow of air in

The larynx sits atop the trachea, composed of horseshoe-shaped cartilages (kraakbeen) that are open at the back and control airflow to the lungs. Inside the larynx, the glottis is the opening between elastic vocal folds, which vary in tension, thickness, and length to produce voiced or voiceless sounds. Phonation follows the myoelastic-aerodynamic theory: subglottal air pressure pushes the folds apart, while the Bernoulli principle causes a pressure drop that sucks them back together, creating rapid vibrations.

Slide 10: • During production the vocal folds are separated for voiceless speech sounds, such as /s,t/, closed for voiced sounds /a, u,

During speech production, vocal folds are separated for voiceless sounds like /s/ and /t/, while they remain closed for voiced vowels such as /a/ and /u/. For voiced consonants like /z/ and /d/, the folds are brought together less firmly to allow voicing alongside high oral air pressure. Additionally, turbulent sounds (e.g., /s/, /f/) result from breath passing through a narrow constriction with an open glottis, whereas transient sounds (e.g., /p/, /t/, /k/) arise from the sudden release of built-up air pressure.

Slide 11: 11

Slide 11 outlines four distinct glottis positions in speech production based on P. Roach's work. The vocal folds remain wide apart for normal breathing and voiceless consonants, while a narrow closure produces whispered vowels like /ahaha/. Voiced sounds require the folds to open and close rapidly to create vibration, whereas a tightly closed position generates a glottal stop.

Slide 12: The fundamental frequency (F0) is the number of times the vocal folds open and close per second. Speech is a complex signal: consists of F0 and

The fundamental frequency (F0) represents the number of times vocal folds open and close per second, measured in Hertz (Hz). Speech is a complex signal consisting of F0 and its multiples, known as harmonics. F0 varies constantly during speech to create intonation and is determined by the elasticity, tension, and mass of the vocal folds; generally, more massive folds vibrate at lower frequencies while tense, elastic folds vibrate at higher frequencies.

Slide 13: Increased vocal intensity is due to greater resistance (by the vocal folds) against the increased airflow. The vocal folds are blown wider apart,

Increased vocal intensity results from greater resistance by the vocal folds against increased airflow. During higher intensity, the vocal folds are blown wider apart, releasing a larger puff of air that creates sound pressure with greater amplitude. Additionally, the vocal folds move farther apart per vibratory cycle and remain adducted for a larger portion of each cycle.

Slide 14: Production of any (speech) sound involves the movement of an airstream

The production of any speech sound requires an airstream movement, most commonly pulmonic egressive where air is pushed from the lungs through the mouth or nose. Other mechanisms include glottalic (implosives and ejectives) and velaric (clicks), which move air within the mouth rather than using lung pressure. While six theoretical combinations exist, only four are attested in languages: pulmonic egressive (universal), velaric ingressive (e.g., Zulu), glottalic egressive (e.g., Navajo), and glottalic ingressive (e.g., Sindhi).

Slide 15: 15

The acoustic signal radiated from the lips consists of sound elements called phonemes, with most languages containing 15-50 units that combine to produce distinct word forms like "man" and "pan." A cognitive or phonetic map linked to our language determines which acoustic differences are meaningful; for instance, the distinction between "pot" and "tot" represents a phonemic contrast, whereas the difference between "pie" and "phie" does not.

Slide 16: Vowel and consonant phonemes are classified in terms of

Vowel and consonant phonemes are classified by manner of articulation, which describes how the vocal tract restricts airflow; this includes complete stops for plosives, varying constrictions for liquids, fricatives, glides, and vowels, and velum lowering for nasal sounds. Classification also relies on place of articulation, referring to the specific location in the vocal tract, and voicing, defined by the presence or absence of vocal fold vibration. Articulatory phonetics connects these linguistic features to the positions and movements of vocal tract articulators.

Slide 17: Vowels: air flows without constriction from lungs through pharyngeal and oral cavities to the outside world

Vowels are produced when air flows from the lungs through the pharyngeal and oral cavities to the outside world without constriction. The vocal tract, extending from the lips to the glottis, serves as the conduit for these sounds, which can be articulated in various locations depending on the language, such as the pharynx or the oral cavity. This unrestricted airflow distinguishes vowels from other speech sounds that involve obstruction or closure within the vocal tract.

Slide 18: Fricatives: narrow constriction in the oral tract (for some languages in the pharynx and in the glottis). If the pressure behind the constriction is high

Fricatives involve a narrow constriction in the oral tract, pharynx, or glottis where high pressure and a sufficiently narrow passage generate turbulent airflow. This manner of articulation includes labiodental (/f, v/), alveolar (/s, z/), dental, velar, uvular, and pharyngeal sounds, as well as the voiceless glottal fricative /h/, which notably lacks constriction. Additionally, affricates are defined as a combination of a stop followed by a fricative.

Slide 19: • Sonorants: continuous, intense, periodic speech sounds: dipthongs, glides, liquids, nasals

Sonorants are defined as continuous, intense, periodic speech sounds that include diphthongs, glides, liquids, and nasals. In contrast, obstruents are weak and aperiodic sounds comprising stop consonants and fricatives, where unvoiced variants are tense and voiced variants are lax due to muscle tension differences. While vowels, liquids, nasals, and fricatives maintain a steady-state articulatory position, stops and glides are transient phonemes involving rapid sequences of closure and release or transition into ensuing vowels.

Slide 20: While manner of articulation and voicing partition phonemes into the broad categories used by most languages, place of articulation (= point of

While manner of articulation and voicing create broad phoneme categories, place of articulation-the point of narrowest vocal tract constriction-enables finer discrimination. Languages vary significantly in the number and choice of these places; for instance, vowel counts range from 3 in Hawaiian to 15 in French, and stop consonant places range from 3 in English to 6 in other languages. Specific consonant places include bilabial (/p/, /b/), labiodental (/f/, /v/), dental, interdental, alveolar (/d/, /t/), palatal, retroflex, velar, uvular, pharyngeal, and glottal.

Slide 21: 21

Slide 21 introduces the concept of "Place of Articulation" as a key parameter in phonetics. This section serves as a heading to categorize where speech sounds are produced within the vocal tract, though no specific definitions or examples are provided in the accompanying source material for this slide.

Slide 22: Vowels are primarily described in terms of

Vowels are primarily described by tongue position relative to the neutral schwa (high/low), the specific part of the tongue raised or lowered (front, middle, back), and lip configuration (rounded/unrounded). Specific vowel types correspond to distinct constriction sites: high front vowels at the hard palate, high back at the soft palate, and low vowels in the pharynx. Additionally, vowels are classified as tense (longer, extreme articulation) or lax (shorter, less extreme) based on duration and displacement from the central schwa position.

Slide 23: General

The International Phonetic Alphabet (IPA) was developed by the International Phonetic Association to provide an unambiguous, one-symbol-per-sound system for representing speech across all languages, addressing inconsistencies in standard spelling. Founded in Paris in 1886 and standardized by 1897, the IPA extends the Roman alphabet with additional symbols to facilitate dictionary pronunciation guides, language recording, and automatic speech recognition transcription.

Slide 24: Phonetic transcription is placed between square brackets [ ] and there is not capitalization or punctuation. An overview of the different characters a

Phonetic transcription uses square brackets without capitalization or punctuation to represent linguistically relevant sounds, excluding personal voice quality. The International Phonetic Alphabet (IPA) classifies consonants by voicing, manner, and place of articulation, while vowels are categorized by features like openness and frontness relative to cardinal vowels. Additionally, the IPA includes symbols for suprasegmental information and non-pulmonic sounds such as clicks, implosives, and ejectives.

Slide 25: 25

The International Phonetic Alphabet organizes symbols by articulatory properties. Consonant charts group places and manners of articulation, while vowel charts use tongue height and frontness/backness, with rounding often indicated separately.

Slide 26: Symbols are foreseen for suprasegmental information, e.g.

Symbols represent suprasegmental information like intonation groups and tone distinctions in languages such as Chinese and Thai, using either iconic lines or non-iconic diacritics to mark pitch height. Diacritics further refine transcriptions by indicating features like aspiration, vowel centralization, breathy voice, and nasalization. Additionally, speech sounds are categorized into broad (phonemic) and narrow (detailed phonetic) transcriptions, both requiring linguistic knowledge to resolve segmentation ambiguities as automatic analysis remains unfeasible.

Slide 27: Experimenta

Slide 27 presents the "Experimenta" vowel and diphthong inventory, organized by tongue position into Back, Mid, and Front categories. The slide lists specific phonemes with example words, including vowels such as /U/ (boot), /ɑ/ (bach), and /i/ (beet), alongside diphthongs like /aɪ/ (buy) and /aʊ/ (down). No additional professor explanation accompanies this visual chart.

Slide 28: Experimenta

Slide 28 categorizes English consonants into Plosives, Fricatives, Affricates, Nasals, and Semivowels, distinguishing them by voicing. The slide lists specific phonemes with example words, such as voiced plosives /b/, /d/, /g/ (boot, dye, goat) and unvoiced fricatives like /f/, /s/, /ʃ/ (fix, sat, ship). It also includes nasals (/m/, /n/, /ŋ/) and semivowels (/w/, /j/) alongside their corresponding examples.

Slide 29: • no one-to-one correspondence between physical sound and phoneme, especially if

There is no one-to-one correspondence between physical sound and phoneme, particularly when articulation is sloppy. The International Phonetic Alphabet (IPA) addresses this by designating a single symbol for the same sound regardless of spelling, though IPA symbols sometimes coincide with orthographic ones. Ultimately, the IPA system is largely based on articulatory properties rather than written conventions.

Slide 30: • Write the symbols for the vowels in the following words

Slide 30 presents phonetic transcription exercises requiring students to write vowel symbols for words like "bread," "rough," and "foot." Additional tasks involve transcribing full words such as "bake" and "bought," with a note that phonemic transcription is acceptable if aspiration is included. Students must also transcribe the sentence, "Opening the bottle presented no difficulty," as part of Exercise 30.

Slide 31: 31

Parametric analyses indicate that every language utilizes a finite set of phonemes to distinguish word meanings. Phoneme variation occurs across dialects, such as the distinction between /u/ and /ʊ/ in most English varieties which is absent in Scots English. In contrast, phonetic variation operates at the individual speaker level, resulting from anatomical differences or specific speaking styles like increased nasality.

Slide 32: Phonology is concerned with

Phonology concerns which speech sounds occur in a language and how they pattern according to regular rules, distinguishing it from the physical production of speech. Phonological knowledge enables speakers to produce meaningful utterances, recognize foreign accents, create new words, and apply grammatical rules like pluralization or aspiration. This system allows individuals to distinguish permissible forms, such as nonsense words that sound like real words, from impossible sound sequences that violate language-specific constraints.

Slide 33: Substitution of glottal stop [ʔ] for [t] does not change the meaning of the word. Therefore, the glottal stop is not a phoneme (it is not a distinctive sound).

Substituting a glottal stop [ʔ] for [t], as in "buʔer" versus "butter," does not alter word meaning, proving the glottal stop is not a distinctive phoneme in English. Because these sounds are interchangeable without creating minimal pairs, they exist in free variation and can be transcribed phonetically with diacritics but lack phonemic distinctiveness. This principle similarly applies to unreleased or released plosives at word endings.

Slide 34: 34

The vowel feature matrix represents vowel categories using phonological features such as height, frontness/backness, and rounding. These are articulatory or contrastive descriptions; acoustic formants are measurable resonances and should not be treated as the same feature system.

Slide 35: 35

In English, the velum automatically lowers to produce nasalized vowels before [+nasal] consonants within the same syllable, while oral vowels occur in final positions or before [-nasal] consonants. This process is governed by a predictable rule: any vowel or diphthong preceding a nasal consonant becomes nasalized. Consequently, nasalized vowels never appear before oral consonants, as demonstrated by the contrast between words like "bee" and "bean."

Slide 36: Redundant versus distinctive

A redundant feature is predictable by rule and thus non-distinctive for a specific segment class, whereas a distinctive feature contrasts meaningfully. In English, nasality is redundant for vowels but distinctive for consonants, illustrating that phonemes lack one-to-one correspondence with phonetic segments. Additionally, sounds in complementary distribution never occur in the same context, such as the velar nasal and glottal glide [h] in English.

Slide 37: Aspiration in English is a redundant feature (not distinctive)

In English, aspiration is a redundant feature because aspirated and nonaspirated voiceless stops (/p, t, k/) occur in complementary distribution, appearing predictably at the start of stressed syllables. Unlike Thai, where aspiration distinguishes meaning, English speakers do not use it distinctively. Similarly, consonant length and specific vowel rounding are predictable in English, whereas they function as distinctive features in languages like Korean, Italian, and Burmese.

Slide 38: For certain classes of sounds the values of some features are universally implied for all languages: all stops [-continuant] are universally and

Certain sound classes universally imply specific feature values across all languages; for instance, all stops ([-continuant]) are predictably [-syllabic]. Speakers possess innate knowledge distinguishing permissible phonemic sequences from impossible ones, as demonstrated by the contrast between acceptable non-words like "blick" and unacceptable strings like "bkli." While specific sound inventories vary, all languages operate under similar phonological rules governing which sequences may occur.

Slide 39: • Phonemes are individual sounds, the smallest meaningful contrastive unit in the phonology of language. Each language has 20-

Phonemes are the smallest meaningful contrastive units in a language's phonology, with each language containing 20-40 of them. Unlike phones, which are the variable physical sounds produced during articulation and written in brackets [ ], phonemes are defined by their function in differentiating words rather than acoustic properties and are written between slashes / /. A single phoneme can be realized as multiple phones called allophones, which are systematic, rule-governed variants such as aspirated or unaspirated stop consonants.

Slide 40: Lack of naturalness results from

Lack of naturalness in speech synthesis results from using too few allophones, failing to distinguish variations like aspirated "kh" versus unaspirated or palatalized "k" and "g" based on stress, vowel type, and syllable position. The auditory system is highly sensitive to these unnatural prosodic patterns, making the accurate implementation of distinct allophonic rules essential for perceived naturalness.

Slide 41: • Syllables are the next larger unit of speech after the phoneme. Usually vowel (= syllable nucleus) and one or more

Syllables are linguistic units larger than phonemes, typically consisting of a vowel nucleus and surrounding consonants. Every utterance contains at least one syllable, which in English can range from a single vowel to complex combinations like "strengths." While people often recognize syllables more readily than individual segments due to writing systems, the average speaking rate produces two to five syllables per second.

Key points to remember

The session further explores the classification of speech sounds using parameters such as manner, place, and voicing, alongside the International Phonetic Alphabet (IPA) for unambiguous transcription. A critical distinction is made between phonemes, the smallest meaningful contrastive units in a language, and phones, their physical realizations or allophones. The lecture emphasizes that many phonetic features, such as aspiration in English or nasalization before nasal consonants, are redundant and predictable by rule rather than distinctive, unlike in languages like Thai where these features alter meaning. Understanding these systematic variations and the rules governing permissible sound sequences is essential for distinguishing natural speech from synthetic output and for accurate linguistic analysis.